A Comparative Evaluation of Encoder and Decoder Language Models with Parameter-Efficient Fine-Tuning (PEFT/LoRA) for Multi-Class Text Emotion Recognition

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***Abstract*—Text-based emotion recognition remains a challenge in Natural Language Processing (NLP) due to severe class imbalance, contextual ambiguity, and informal language in short digital texts. While traditional encoder-based architectures and recurrent neural networks have served as standard baselines, modern generative Large Language Models (LLMs) offer promising capabilities for capturing subtle affective nuances. However, full fine-tuning of these models remains computationally prohibitive. In this work, we present an empirical evaluation comparing traditional encoder models (BiLSTM, BERT) against intermediate-scale decoder architectures (Qwen, Gemma, DeepSeek) using Parameter-Efficient Fine-Tuning (PEFT/LoRA). Experiments on the CARER / Kaggle Emotion dataset demonstrate the trade-offs in classification performance, measure via Macro-F1, under controlled hardware constraints.**

***Index Terms*—Text Emotion Recognition, Large Language Models, LoRA, PEFT, Macro-F1, Multi-class Classification.**

I. Introduction

Text-based emotion recognition is central to applications ranging from conversational agents to consumer feedback analysis and mental health monitoring. However, reliably detecting fine-grained emotional states in brief, informal digital texts remains difficult due to contextual ambiguity and severe class imbalance. While recurrent models like BiLSTM and encoder-based architectures like BERT have long served as primary baselines, generative Large Language Models (LLMs) have demonstrated strong potential in capturing complex semantic variations through advanced pre-training. Despite this potential, the massive parameter size of modern decoders makes full fine-tuning computationally unfeasible for resource-constrained environments, highlighting the need for efficient adaptation techniques.

Parameter-Efficient Fine-Tuning (PEFT) methods, particularly Low-Rank Adaptation (LoRA), address these hardware bottlenecks by freezing the pre-trained base model and updating only a small subset of low-rank adapter weights. Although decoders such as Qwen, Gemma, and DeepSeek excel in general language tasks, their performance when adapted via LoRA for specialized downstream classification under high class imbalance is not yet fully understood. Existing benchmarks often focus on broad sentiment analysis or general classification, relying on overall accuracy, a metric that frequently masks poor performance on minority emotional classes. Furthermore, direct comparisons between fully fine-tuned lightweight encoders and LoRA-adapted intermediate decoders under identical hardware constraints remain scarce in current literature.

To address these gaps, this work evaluates the trade-offs between legacy encoder baselines and modern intermediate-scale decoders on the highly imbalanced CARER emotion dataset.

Our main contributions are as follows:

- **Cross-Architecture Benchmark:** We provide a systematic empirical comparison between state-of-the-art intermediate decoders (Qwen, Gemma, DeepSeek) and established encoder baselines (BiLSTM, BERT) for multi-class emotion classification.
- **Efficiency & Optimization Analysis:** We evaluate the operational trade-offs of PEFT/LoRA across decoder architectures, focusing on memory footprint, parameter count, and convergence rates under strict hardware limitations.
- **Imbalance-Aware Evaluation:** We analyze model resilience to severe class imbalance on the CARER dataset, prioritizing Macro-F1 metrics to ensure fair assessment across underrepresented emotion classes.
- **Practical Deployment Guidelines:** We outline concrete performance-versus-compute trade-offs to offer actionable decision frameworks for researchers and practitioners choosing between compact encoders and LoRA-adapted decoders.

The remainder of this survey is structured as follows: Section II reviews the evolution and current approaches in text-based emotion recognition. Section III analyzes parameter-efficient adaptation methods across Encoder and Decoder architectures. Section IV examines the benchmark dataset characteristics and evaluation protocols. Section V outlines our proposed experimental setup and methodology for future empirical training. Section VI discusses preliminary trade-offs and future research directions, and Section VII concludes the paper.

II. Domain Analysis & Evolution of Text Emotion Recognition

A. Dominant Traditional Approaches

Text-based emotion recognition has evolved through a rapid progression of computational methodologies, which are designed to address the challenges related to semantic ambiguity and unstructured language. Some of these computational approaches have relied on multiple computational paradigms to interpret a text and determine which emotions are shown. These foundational techniques can be broadly categorized into keyword-based, corpus-based, ruled-based, tradicional machine learning and deep learning.

- **Keyword-Based Approaches**: Identify explicit emotional terms using lexical databases (such as WordNet-Affect or SentiWordNet) and apply linguistic rules to assess sentiment intensity [**?**].
- **Corpus-Based Approaches**: Induce domain-specific word-emotion lexicons using statistical methods and generative models across annotated or weakly-labeled text datasets [**?**].
- **Rule-Based Approaches**: Apply expert heuristic rules, Part-of-Speech tagging, and phrasal patterns to infer emotional states based on predefined linguistic structure [**?**].
- **Traditional Machine Learning Approaches**: Utilize supervised algorithms, such as, Support Vector Machines (SVM), Naïve Bayes, Decision Trees, and K-Nearest Neighbors—that rely on manual feature engineering to classify text into emotional categories [**?**].
- **Deep Learning Approaches**: Employ neural network architectures, such as, Recurrent Neural Networks (RNNs/LSTMs), Convolutional Neural Networks (CNNs), and Transformer-based models (e.g., BERT), to automatically learn hierarchical representations and capture complex contextual semantics directly from raw text[1].

B. The Shift to Generative Decoders

The rise of generative Large Language Models (LLMs) has fundamentally changed how we approach text classification in NLP [1]. Traditionally, affective computing tasks relied on discriminative models like BERT or GRU variants that learn a direct mapping function $f(X) \to Y$ from input text X to a predefined set of labels Y [2]. In contrast, autoregressive decoder architectures treat classification as a conditional text generation problem. They model the probability of generating a target class token y given a prompt sequence X:

$$P(y \mid X) = \prod_{i=1}^{N} P(w_i \mid w_1, \ldots, w_{i-1}, X) \tag{1}$$

This shift offers significant benefits, especially when it comes to leveraging broad contextual knowledge and picking up on subtle emotional tone that standard encoder representations often miss [3].

1) Zero-Shot and Few-Shot Prompting Paradigms: A key feature of generative decoders is their ability to handle downstream classification without updating model weights, relying instead on in-context learning (ICL) [4]. In a zero-shot setup, the model gets an instruction template describing the task along with the input text, using its pre-trained knowledge to output the predicted label. Few-shot prompting builds on this by adding k example pairs directly into the prompt context:

$$\text{Prompt} = \{(X_1, Y_1), (X_2, Y_2), \ldots, (X_k, Y_k), X_{\text{test}}\} \tag{2}$$

While zero-shot and few-shot setups avoid task-specific training costs, their reliability for fine-grained emotion tasks varies greatly. Zero-shot decoders are notoriously sensitive to how prompts are written, how context examples are ordered, and how subtle affective context is presented. Furthermore, without parameter adaptation for a specific domain, zero-shot LLMs often struggle with subtle emotion boundaries.

2) Supervised Fine-Tuning versus In-Context Learning: To address the instability of in-context learning, supervised fine-tuning (SFT) adjusts the model's output probabilities to align directly with task-specific labels. In multi-class emotion recognition, SFT restricts the output space to valid emotion categories while teaching the model to handle informal digital slang and implicit emotional cues [5].

However, choosing between prompting and full fine-tuning presents clear trade-offs:

- **In-Context Learning (Zero/Few-Shot):** Keeps base weights frozen and avoids training costs, but can struggle with subtle emotional nuances.
- **Full Supervised Fine-Tuning (SFT):** Delivers strong task alignment across imbalanced emotion classes, but requires updating billions of parameters ($> 10^9$), creating massive computational and memory bottlenecks during backpropagation.

This balance demonstrates why intermediate approaches are necessary. Generative decoders provide strong semantic understanding, but using them effectively for emotion classification requires parameter-efficient strategies, such as Low-Rank Adaptation (LoRA), that deliver SFT performance without excessive computational demands [6].

C. Established Standards and Open Challenges

Sentiment analysis and emotion recognition overlap significantly in computational linguistics, but they present very different levels of complexity. Distinguishing basic sentiment classification from fine-grained emotion recognition makes it easier to pinpoint the key challenges facing modern NLP.

1) Established Baselines in Binary Sentiment Analysis: Binary sentiment classification, which sorts text into simple positive or negative categories, is a well-established task. Earlier studies applying fine-tuned Encoders like BERT alongside GRU variants reached near-peak performance on benchmarks such as SST-2 and IMDb [2]. Because these datasets rely on balanced classes and explicit sentiment words, standard supervised models handle well-structured text with high reliability.

2) Open Challenge 1: Fine-Grained Multi-Class Emotion Mapping: Moving beyond binary sentiment requires mapping text to specific psychological states, such as Ekman's six core emotions: anger, fear, joy, love, sadness, and surprise. The primary difficulty here is overlapping class boundaries. Distinct emotions like fear and surprise frequently share identical keywords, meaning the model must rely on broader context rather than individual terms to tell them apart [5]. Modern decoder architectures capture these subtle contextual shifts well, but mapping their outputs into discrete emotion categories without misclassifications remains difficult [4].

3) Open Challenge 2: Extreme Class Imbalance in Affective Corpora: A major bottleneck in emotion datasets like the CARER benchmark is severe class imbalance [3]. Everyday language features common emotions like joy or sadness far more often than high-intensity states like surprise or fear. Standard cross-entropy loss naturally focuses on majority-class errors, which leads to high overall accuracy but poor performance on minority categories. While sampling methods and weighted losses help traditional encoders, managing severe imbalance within Parameter-Efficient Fine-Tuning (PEFT) frameworks, where base weights remain frozen, is still an ongoing challenge for Macro-F1 optimization [6].

4) Open Challenge 3: Ambiguity in Digital Text: Short online texts introduce noise that degrades classification accuracy. The main causes of ambiguity include:

- **Implicit Emotions and Sarcasm:** Phrases without direct emotion keywords (such as *"Great, another delay."*) depend on pragmatic reasoning that basic encoders often miss.
- **Informal Language:** Heavy use of slang, shortcuts, irregular grammar, and repeated punctuation (like *"???"*) creates out-of-vocabulary problems during tokenization.
- **Limited Context:** Brief posts provide very little surrounding text, making it harder to determine the exact emotion of words that change meaning based on subtext.

Balancing small, efficient architectures against the deep contextual understanding needed to resolve these ambiguities within strict memory limits remains a major objective for practical emotion recognition systems.

III. Parameter-Efficient Adaptation of Encoder and Decoder Architectures

A. Full Fine-Tuning vs. PEFT Paradigms

B. Mechanics of Low-Rank Adaptation (LoRA)

C. Comparative Methodologies: LoRA vs. QLoRA vs. Full Fine-Tuning

D. Theoretical Limitations and Underlying Assumptions

IV. Benchmark Analysis: The CARER Dataset & Evaluation Protocols

A. Corpus Characteristics and Domain Scope

B. Reported Metrics and Published Baselines

C. Known Dataset Limitations and Biases

V. Proposed Experimental Methodology

A. Experimental Framework and Hardware Constraints

B. Model Selection and Adapter Configuration

C. Pre-Processing and Imbalance Mitigation Strategy

D. Evaluation Protocol and Statistical Validation

VI. Discussion & Open Research Directions

VII. Conclusion

References

[1] P. Chakriswaran, D. R. Vincent, K. Srinivasan, V. Sharma, C.-Y. Chang, and D. G. Reina, "Emotion ai-driven sentiment analysis: A survey, future research directions, and open issues," *Applied Sciences*, vol. 9, no. 24, p. 5462, 2019. [Online]. Available: https://www.mdpi.com/2076-3417/9/24/5462

[2] C. Zhang, "A comparative exploration of bert, rnn, and gru for sentiment classification," *Highlights in Science, Engineering and Technology*, vol. 120, pp. 54–58, Dec. 2024. [Online]. Available: https://drpress.org/ojs/index.php/HSET/article/view/28777

[3] E. Saravia, H.-C. T. Liu, Y.-H. Huang, J. Wu, and Y.-S. Chen, "CARER: Contextualized affect representations for emotion recognition," in *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, E. Riloff, D. Chiang, J. Hockenmaier, and J. Tsujii, Eds. Brussels, Belgium: Association for Computational Linguistics, Oct.-Nov. 2018, pp. 3687–3697. [Online]. Available: https://aclanthology.org/D18-1404/

[4] Y. Cheng, H. Chen, X. Guo, B. Ding, and Z. Pei, "Personality-aware emotion recognition in conversation with large language models," *Pattern Recognition*, vol. 178, p. 113460, 2026. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0031320326004267

[5] B. Kratzwald, S. Ilić, M. Kraus, S. Feuerriegel, and H. Prendinger, "Deep learning for affective computing: Text-based emotion recognition in decision support," *Decision Support Systems*, vol. 115, pp. 24–35, 2018. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0167923618301519

[6] S. Nwaiwu, "Parameter-efficient fine-tuning for low-resource text classification: a comparative study of lora, ia3, and reft," *Frontiers in Big Data*, vol. 8, 2025. [Online]. Available: https://www.frontiersin.org/journals/big-data/articles/10.3389/fdata.2025.1677331

[7] A. Mohammed and R. Kora, "Enhancing text-based deepfake detection: A comprehensive evaluation of lora fine-tuned large language models," *IEEE Access*, vol. 14, pp. 44 709–44 727, 2026.

TABLE I
Comparative Summary of Existing Approaches in Affective Computing and PEFT Architectures

Reference	**Year**	**Technique**	**Dataset / Environment**	**Primary Metric**	**Reported Result**	**Identified Limitation**
Saravia et al. [3]	2018	BiLSTM + Contextual Word Embeddings	CARER (Emotion Benchmark)	Macro-F1	0.647 (Benchmark Baseline)	Limited long-range semantic capacity; struggles with severe class imbalance.
Kratzwald et al. [5]	2018	Deep Learning (BiLSTM / CNN)	Financial & Customer Reviews	Classification Accuracy	82.4% Accuracy	High dependence on task-specific fine-tuning; poor zero-shot generalization.
Chakriswaran et al. [1]	2019	Lexicon-Based & Deep Learning Survey	Multimodal / Microblogs	Accuracy / F1-Score	Qualitative SOTA Overview	Highlights systemic issues with informal slang, sarcasm, and class skew.
Zhang [2]	2024	BERT, RNN, GRU Full Fine-Tuning	Standard Sentiment Corpora	Accuracy / Macro-F1	BERT outperforms RNN/GRU ($> 90\%$)	Full fine-tuning requires substantial memory; limited to fixed context lengths.
Nwaiwu [6]	2025	LoRA, $(IA)^3$, ReFT Evaluation	Low-Resource Text Benchmarks	Macro-F1 / Memory Footprint	LoRA achieves $\sim 98\%$ of full fine-tuning	Performance varies significantly across adapter target modules and target tasks.
Mohammed & Kora [7]	2026	LLMs + LoRA (PEFT)	Text-Based Deepfake Corpora	Detection Accuracy / F1	High parameter efficiency ($> 92\%$)	Performance drops under extreme quantization or overly restricted rank (r).
Cheng et al. [4]	2026	LLMs In-Context Learning / SFT	Conversational Emotion Datasets	Macro-F1	Improved multi-turn emotion tracking	Highly sensitive to prompt context; expensive inference latency for broad LLMs.